Reg. No									

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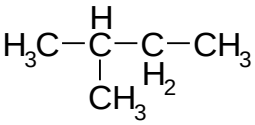
CH101S – ORGANIC CHEMISTRY - I

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer



1. According to IUPAC nomenclature, is known as
   a) 2-methylbutane b) 3-methybutane
   c) 2-methylpentane d) 3-methylpentane

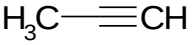
2. Inductive effect of alkyl groups is in the order.
   a) Tertiary>secondary>primary b) Primary>secondary>tertiary
   c) Secondary>primary>tertiary d) Secondary>tertiary>primary

3. Cyclic ketones may be converted into cyclic alkanesby
   a) Clemenson's reduction b) Wustz reaction
   c) Dickmann reaction d) None of the above

4. Cydopropane with Br_2 in dark gives
   a) Dibromocyclopropare b) 1,3-dibranopropane
   c)1,2-dibranopropare d) Monobramocyclopropare

5. Types/Type of isomesism exbibited by alkenes.
   a) Chair isomesism b) Position isomesism
   c) Geometrical isomesism d) All of the above

6. Which of the following halides with alkene
   a) Hcl b) H-I c) H-Br d) None of the above

7. IUPAC name for  is
   a) Propyne b) Allylene
   c) Methylacetylere d) Acetylmethyne

8. The catalyst used is hydration of alkynes is
   a) H_2SO_4 b) $HgSO_4$ c) Cu d) Pd

9. The most syable contosmer of n-butane is
   a) Syaggered b) Eclipsed
   c) Anti-syaggered d) Fully eclipsed

10. In trans-1,2-dimethylcyclohexane, methylgroups are present is __ positions
    a) ae b) ea c) aa d) aa or ee

II. Fill up the blanks

11. The type of hybridization involued in acetylene is _____

12. The metal used in wastz reaction is _____

13. In the presence of peroxide propylene reacts with HBr and yields_____

14. The untable product formed when acetylere reacts with water is ___

15. The melting point of isomer is greater than c is isomer. In is became of_____

III. Match the following

16. SP^2 hybridisation - 1,3-butadiene

17. Baeyer Strain theory - Syn and anti

18. Saytzeff's rule - Quatenary ammonium salt

19. Conjugated diene - Lower cyclo alkanes

20. Ketoximes - Ethylene

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Define bond angle.

22. What is crductive effect?

23. What are casbarions? Give an example.

24. Give Rolbe's method of preparation of alkanes.

25. Give any two substitution reactions of cycloalkanes.

26. Define markovnikov's rule in alkene addition.

27. Write the polymerization reaction of ethylene.

28. Write the addition reaction of unsymmetrical alkynes to hydrogen halides.

29. What is Diels Alder reaction.

30. What are conformers.

31. Give E/Z nomenclature of maleic and formaric acids.

32. Draw the fully eclipsed and antistaggered conformess of n-butane.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Explain hyperconjugation with an example. **(4)**
    b) Explain the cleavage of bonds. **(3)**

34. a) Outline the preparation of alkenes by kolbe's method. **(3)**
    b) Explain Baeyer's strain theory. **(4)**

35. a) Give Dieckmann's king closme reaction for the preparation of cycloalkanes. **(3)**
    b) Write down any four chemical reactions of alkanes. **(4)**

36. a) Explain saytzeff's rule with an example. **(4)**
    b) Explain Anti Markovnuikov's rule with an example. **(3)**

37. a) Explain 1,2 and 1,4 –additions in diens with examples. **(4)**
    b) Give examples foir conjugated, isolated and cumulative dienes. **(3)**

38. a) Draw the chair form of cyclohexane with axial and equitorian hydrogens. **(3)**
    b) Explain the conformational analys is of n-butane. **(4)**

39. a) Give any two methods of distinguishing geometrical isomers. **(4)**
    b) Define conformens and torsional strain. **(3)**

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